



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Adaptive Software Engineering – (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Based Smart Food Redistribution and Waste Reduction System	
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Abstract

The AI-Based Smart Food Redistribution and Waste Reduction System is an intelligent platform developed to reduce food waste and ensure surplus food reaches people in need. It connects food donors such as restaurants, hotels, supermarkets, and households with NGOs, food banks, and charitable organizations through a centralized digital platform. The system uses Artificial Intelligence (AI) to analyze food availability, predict demand, prioritize donations based on freshness and expiry time, and recommend the most efficient distribution routes. It also provides real-time inventory management, donation tracking, automated notifications, and secure user authentication to improve transparency and efficiency. The platform enables quick coordination between donors, receivers, and delivery partners, ensuring timely food redistribution while minimizing spoilage. By reducing unnecessary food waste, the system supports environmental sustainability, lowers carbon emissions, and promotes social welfare by addressing food insecurity. The application is built using modern technologies such as React.js for the frontend, Python (FastAPI) for the backend, PostgreSQL for database management, TensorFlow for AI-based prediction, REST APIs for communication, JWT Authentication for security, and Google Maps API for route optimization. The system also uses Docker for containerization, Kubernetes for container orchestration, and cloud technologies for scalable deployment and service management. It follows Adaptive Software Engineering, AI, Cloud-Native, and Service-Oriented Architecture concepts to support flexible development, intelligent processing, and modular services. This solution offers a scalable, secure, and technology-driven approach to sustainable food management.

Signature of the Guide

HOD